

REVIEW

Bacterial Control of Plant Diseases

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This article focuses on the effective biocontrol of plant diseases by microorganisms, which is attracting attention as an alternative to chemical control methods. As most research has so far been concentrated on fluorescent *Pseudomonas* species, the use of *Bacillus* species which has been considered to be less effective compared to that of pseudomonads, has been mainly introduced to demonstrate the effectiveness of the bacteria.

[Key words: biocontrol, *Bacillus subtilis*, antifungal substances, solid state cultivation]

The overuse of chemical pesticides to cure or prevent plant diseases has caused soil pollution and had harmful effects on human beings. Accordingly, to reduce the use of these chemicals, one possibility is to utilize the activity of microorganisms. It is desirable to replace chemical pesticides with materials that possess the following three criteria: high specificity against the targeted plant pathogens, easy degradability after effective usage, and low cost of mass production. Products produced biologically or the microbial cells themselves are called biological control (biocontrol) agents or biological pesticides if they fulfill these criteria. It is ideal if the microorganisms that produce such products stably inhabit the environment as nondominant species but maintain their effectiveness.

Under these circumstances, biocontrol of some soil-borne diseases has been attracting attention recently. As most of the soil-borne plant pathogens are fungi, biocontrol by fungi has been attempted intensively (1–5). The use of bacteria has also been investigated mainly because genetic and biochemical analyses and the mass production of bacteria or bacterial products are much easier than those of fungi, and thus the advance of bacterial control is expected to have great potential. As bacterial control agents, *Agrobacterium*, *Pseudomonas*, *Bacillus*, *Alcaligenes*, *Streptomyces*, and others have been reported. Different mechanisms are involved in the actions of these microorganisms against plant pathogens, such as parasitism, cross protection, antibiosis, and competition. Parasitism is the dependence of one species on the pathogens as the host, which is commonly observed among fungi. For example, a fungus invades the mycelium of a fungal pathogen by producing β -(1, 3) glucanase and chitinase, causing the death of the plant pathogen. When a nonpathogenic strain or a weakly-pathogenic strain is crossed with pathogens, the occurrence of disease is significantly reduced. This phenomenon is defined as cross protection. One example of a nonpathogenic *Fusarium oxysporium* crossed with the pathogen *F. oxysporium* f. sp. *batatas* is reported in the case of stem rot of sweet potato (6), but few cases are reported in relation to bacterial cross protection. The antibiotic mechanism is said to operate when the metabolic products (antibiotics) produced by one species inhibit or suppress the growth of another species. Competition

involving nutritional or spatial struggle occurs among the microorganisms. It is critical to note in relation to biocontrol that while inhibition of the growth of plant pathogens is observed *in vitro*, often no correlation is observed between inhibition on plates and reduction of diseases in a test in greenhouse or in a field. Therefore, a plant test is essential to confirm the effectiveness of biocontrol agents.

In this review, bacterial control of plant pathogens which cause soil-borne and air-borne diseases was discussed mainly from the biochemical and engineering viewpoints. As pseudomonads already have a long history as a candidate of bacterial control agent and some reviews and reports of intensive research have been published (7–11), recent advances in the use of *Bacillus* spp. are emphasized.

BIOLOGICAL CONTROL OF PLANT PATHOGENS BY *PSEUDOMONAS* SPECIES

Pseudomonas strains are one of the most active and dominant bacteria in the rhizosphere and have been intensively investigated as biocontrol agents (12–15). Among them, *P. fluorescens*, *P. putida* and *P. cepacia* were the predominant focus of research for practical applications. The strains of *Pseudomonas* produce several kinds of antibiotics, such as pyrolnitrin, pyoluteorin and phenazine-1-carboxylate, all of which are closely related to the suppression of plant diseases (7, 9–23).

Siderophores are extracellular, low-molecular-weight compounds which have a high affinity for ferric ions and play a role in the biocontrol (24). Some *Pseudomonas* strains were confirmed to suppress the activities of plant pathogens by the production of siderophore or pseudobactin (25, 26). Bacterial siderophores are also known to have functions to help plant growth such as supplying iron via an iron-chelating function called the PGPR (Plant Growth Promoting Rhizobacteria) effect (24, 27, 28), which eventually leads to a decrease in disease occurrence. A siderophore-producing *P. fluorescens* induced an increase in the emergence of cotton seedlings in a soil infested with *Pythium ultimum*, while siderophore-deficient mutants of *P. fluorescens* derived by Tn5 mutagenesis induced a decrease in the emergence of seedlings

(27). Genetic analysis and regulation of some metabolites (23, 29) and experiments on the ecological fitness of soil (18, 20, 27, 30, 31) have been extensively carried out.

BIOLOGICAL CONTROL OF PLANT PATHOGENS BY *BACILLUS* SPECIES

The use of the gram-positive *Bacillus* species as a biocontrol agent has been relatively rare and has been studied less intensively than that of the gram-negative bacteria. The antagonists studied are mainly *Bacillus subtilis* and occasionally *B. megaterium*, *B. cereus*, *B. pumilus*, and *B. polymyxa* (32–34). As *Bacillus* spp. have the characteristics of omnipresence in soils, high thermal tolerance, rapid growth in liquid culture, ready formation of resistant spores, and are considered to be a safe biological agent, their potential as a biocontrol agent is considered to be high. However, evaluation of the bacteria has focused primarily on the degree of disease suppression (35–37). The population dynamics and mechanism of suppression to plant pathogens in soil by *Bacillus* spp. have not been extensively investigated.

Production of antifungal substances *B. subtilis* produces several kinds of antibiotics, e.g., bacillomycin (38, 39), iturin (40–42), mycosubtilin (43), bacilysin (44, 45), fengymycin (44) and mycobacillin (46). However, the bacteria require a suitable substrate for production in soil. Investigation of the control of *Streptomyces scabies* in potatoes by *B. subtilis*, revealed that more antibiotic was produced when the bacteria were grown a water extract of soybean (47). Furthermore, when this material was added to soil, buildup of potato scab was prevented, presumably because the soybean substrate supported antibiotic production in soil by *B. subtilis* thus enhancing its ability to control the specific disease. A root system includes a wide variety of different substances whose compositions vary with changes in the metabolic state of different parts of the root system. The changes can be influenced by outside effects on the plants. Also, once a root is invaded by a pathogen, significant changes in exudation occur (48). Such changes could convert the environment into one hostile to the biocontrol agent. Because antibiotic activity is so dependent on outside events, biocontrol through its effects is likely to be unsatisfactory and certainly unpredictable, unless strains of microorganisms can be manipulated to make the synthesis of antagonistic compounds less susceptible to changes in nutrient sources.

Production of siderophores 2,3-Dihydroxybenzoyl-glycine (2,3-DHBG) is known as a siderophore produced by the gram-positive *B. subtilis* (24, 49). The strains of *B. subtilis* which showed a wide suppressive spectrum on plant pathogens by producing antibiotics, also produce 2,3-DHBG. Gram-negative *E. coli* produces a siderophore, enterobactin (50); the *E. coli* mutant of *entD* encoding enterobactin was complemented for enterobactin production by the *sfp*⁺ gene of *B. subtilis* (51) which differs from *sfp* (surfactin production gene) by five base substitutions and one base insertion that truncated a 224-amino-acid residue in *sfp* to a 165-amino-acid residue. Similar complementation was found in the regulatory gene, *lpa-14* for the antibiotics iturin A and surfactin derived from a potential bacterial control agent of *B. subtilis* (52–55). The finding that *B. subtilis* produced not only antibiotics which suppress plant pathogens, but also siderophores, and the regulation of

these products by the gene *lpa-14*, indicate the possibility of enhanced effectiveness of biocontrol by the manipulation of the gene.

Genetic analysis Some antifungal peptides produced by *Bacillus* species are synthesized by multienzyme complexes via nonribosomal mechanisms and the molecular organization of these peptide synthetases encoded by the bacterial operons has been well analyzed (56). However, no relation between the genetic information and the suppressive effect on plant pathogens in soil has been reported. Recently, one regulatory gene, *lpa-14*, which is responsible for the production of antibiotics, was found to be closely involved in the bacterial suppressive function against a plant disease in soil (52–55). In a plant test in soil, infestation by the plant pathogen *R. solani* was conducted and the percentage of diseased plants was 85.2%, while the introduction of *B. subtilis* into the soil reduced the percentage of diseased plants to 24.1%. When RΔ1 which destroys of *lpa-14* was inoculated into the soil, the percentage was 64.8%, which was significantly lower than that following control treatment of soil with *R. solani* (85.2%), and the shoot lengths and shoot weights were significantly higher than those in the controls (54). This suggests that other determinants, such as unknown antibiotics, lytic enzymes, or siderophores may be involved in the activity of RΔ1. Similar phenomena were reported for *P. fluorescens* strains which produce siderophores (27) or the antibiotic phenazine-1-carboxylate (18), in that they retained suppressiveness even when mutants lacking the production of these substances were prepared. These data indicate that other unknown functions are operative in effective microorganisms which attack plant pathogens.

Ecological fitness in soil A few investigations have been done from the ecological point of view of *Bacillus* spp. The environmental factors for the production of the antifungal compounds by these spp. antagonistic to plant pathogens are important factors. *B. subtilis* antagonistic to the pathogen of apple crown rot had the optimal temperature of 21–28°C for the production of antifungal compounds, but the most suitable temperature for its growth was 25°C. This antagonistic bacterium increased its population density after 33 d when applied to the presterilized soil. However, as the soil extract did not support the growth of *B. subtilis* or the production of antifungal substances, amendment of nutrients was necessary (57, 58). Another *B. subtilis* strain, showed the optimal temperature for the production of antifungal substances at 30°C in liquid cultivation, but at below 25°C in solid-state cultivation (59, 60). The temperature difference is reflected in the ecological behaviour in soil, mostly in relation to the suppressive capacity of the bacterium as the average soil temperature of mainland Japan is below 25°C.

It is often said that *Bacillus* strains are not normally stimulated in the root region (61). However, when introduced massively into soil (23), they multiply and colonize parts of the roots. They are good competitors and attach themselves firmly to root surfaces, especially if an inoculum of spores is used. A spore inoculum for *Bacillus* strains is advantageous to use in biocontrol preparations because it is resistant to adverse environmental conditions and remains viable for several years under any storage conditions. However, environmental conditions must be appropriate for the spores to germinate to vegetative cells in the soil following seeding.

As the generation time of *Bacillus* species is slower than that of *Pseudomonas* species, *Bacillus* species are less likely to lead to modify the rhizosphere microflora.

Although *B. subtilis* is not considered a representative rhizosphere strain as is *Pseudomonas* spp., the rhizosphere population density as well as the persistence of the bacterium in soil is an important factor in the suppression of plant diseases in soil. *B. subtilis* is uniformly distributed throughout the soil and is not concentrated in the plant rhizosphere. The population is considered to be 10^3 – 10^5 cells/g of soil, indicating that it is a minor species in soil. The threshold population density of bacteria to show significant suppression of plant pathogens was reported as 10^5 cells/g of root in the case of *Pseudomonas* spp. (62). It was found that most of the cells of *B. subtilis* recovered from soil were in the form of spores that had no ability to produce antibiotics, however the obvious reduction of plant disease suggests that the spores germinate in the rhizosphere and the antibiotics are produced; the exact number of vegetative cells was not determined.

As well as survival of microorganisms, persistence of antibiotics produced by bacteria in soil is crucial for the maintenance of their stable efficacy on plant pathogens. Figure 1 shows an example of the stability of the antibiotics iturin A and surfactin in soil into which the solutions of these substances were introduced. Surfactin was more persistent than iturin A in the soil. Decrease in the amount of these antibiotics was caused by leaching from soil during watering, biodegradation by microorganisms or irreversible binding to soil materials or humic acids. Since the concentrations of iturin A and surfactin in a centrifuged culture supernatant were stably maintained for more than 1 month, such instability was related to the soil environment. When vegetative cells of *B. subtilis* were inoculated into the soil, the survival rate of the cells was very low. However, once, the spores of the bacterium were introduced or sporulation from vegetative cells occurred, mainly because of the oligotrophic condition of the soil, the population was stabilized in accordance with the spore number on the order of 10^7 cells/g of dry soil, and almost no change was observed for 30 d (63). Therefore, if the initial populations are well maintained in soil and most of the cells exist as spores, *B. subtilis* is stable in the soil as spores; the spore's stability and ease of handling are advantageous for the use of this bacterium as a biocontrol agent.

Production of antibiotics in liquid cultivation For the mass production of antagonistic bacteria or antibiotics produced by bacteria, liquid cultivation is the conventional method. With the advances in bioreactor-based biotechnological processes, it has become important to develop economical methods for the isolation and purification of products from cultures or cells which are suitable for large-scale application in biological control.

When *B. subtilis*, which has the ability to suppress plant pathogens by producing lipopeptide antibiotics, was cultivated in conventional liquid cultivation, the lipopeptide was condensed only in the foam formed during cultivation (64). This indicates that continuous separation and condensation of the product is possible only by collecting the foam under properly controlled foaming conditions.

Some sophisticated separation methods, such as membrane separation or chromatographic procedures, are successful as laboratory methods but have limitations as

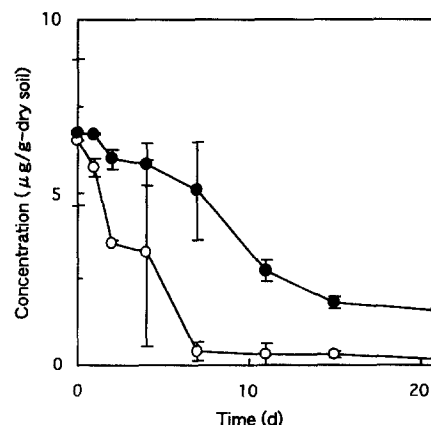


FIG. 1. Stability of lipopeptide antibiotics, iturin A (○) and surfactin (●) which are associated with suppression of plant pathogens in soil. The bars represent standard derivations. The details of soil treatment and analytical procedures are the same as those in reference (54).

large-scale processes. Foam separation is a possible and simpler separation method. Foam separation has been reported for the removal of several microorganisms from aqueous suspensions (65–67) or for the separation of extracellular proteins produced by *Saccharomyces cerevisiae* (68). However, foam separation and the concentrations of peptidolipidic substances in a jar fermentor have not been reported. Foaming in a jar fermentor is generally cumbersome and the use of an antifoam agent is in most cases unavoidable. The feature of this method is closely related to the chemical nature of the product, which contains both hydrophobic and hydrophilic parts in its structure. The advantage of this method also lies in the minimal use of an antifoam agent, which is sometimes causes physiological and economical disadvantage in fermentation. The sample obtained in this procedure was used for a plant test and the effectiveness of it was confirmed (54).

Operation of solid state cultivation In order to produce bacterial control agents, solid-state cultivation of bacteria using solid organic wastes is one possibility. More than 2 million tons of organic waste available for microbial treatment are disposed off annually from the food industry in Japan. However, direct application of these organic matters to soil causes severe damage to plants due to the low ratio of carbon to nitrogen in them. Therefore, some pretreatment before introduction into soil is needed. One of the materials is a soy bean curd residue, which is a by-product of tofu manufacturing. Approximately 700,000 tons of the residue is produced annually; most of which is incinerated as an industrial waste. In the 15 g to 3 kg scale of cultivation of soybean curd residue for the growth of *B. subtilis*, the accumulation of the antibiotics associated with suppression of plant pathogens was confirmed when proper temperature control and air supply were maintained (59).

Table 1 shows a comparison of the production of iturin A between solid-state cultivation and liquid cultivation. Cultivation time in the solid-state cultivation is only 2 d for the maximum production of iturin A, while that in liquid cultivation is 5 d. When the production of iturin A is evaluated per unit gram of wet substrate, solid-state cultivation was 6–8 times more efficient than liquid cultivation. The contents of carbon and nitrogen

TABLE 1. Comparison of the antibiotic, iturin A production between in solid state cultivation and in liquid cultivation

	Solid state cultivation	Liquid cultivation
Scale	3 (kg)	5 (l)
Cultivation time (d)	2	5
Iturin productivity (mg/g wet culture/d)	0.68	0.038
Total iturin (mg) per container	5500	900
Culture volume (l) per 1 g iturin	1.5	7.2

in the water soluble fraction of the soil material was about 21.3 g-C/l solid, and 2.3 g-N/l solid respectively. In the medium in liquid cultivation, the contents of carbon and nitrogen were 17.7 g-C/l medium and 4 g-N/l medium, respectively. Since 1 l of solid was almost equal to 1 kg in weight, a comparison of the two methods is plausible.

As iturin A has five homologues, the production of fractions with a longer side chain which show stronger antibiotic activity was greater in solid state cultivation, indicating that solid-state cultivation has an advantage over liquid cultivation at least in regard to this point (59).

B. subtilis is a good candidate for this treatment and the product treated by the bacterium can be utilized as an organic material which functions both as an organic fertilizer and a microbial pesticide. As these organic matters are vulnerable to decay, a new oven drying process was developed and the processed product exhibited improved transportability and preservability, and was found to be effective for growing the bacterium as a solid substrate to produce the cells and the antibiotics (69).

Solid-state cultivation has the following features.

(i) Traditional oriental foods are produced by solid state cultivation (e.g., enzymes, organic acids, koji-making, etc.); (ii) Most of the microorganisms used are fungi. Little bacterial usage is encountered; (iii) Solid substrates require only addition of water or other nutrients; (iv) Vessels in solid-state cultivation are small relative to the product yield; (v) Low moisture content reduces the problem of contamination; (vi) Extraction processes are simpler; (vii) Amount of solvents used for extraction is much smaller; (viii) Spore inoculation does not need a seed tank; (ix) Products can be used for direct applications such as animal feed, fertilizer etc.

The extraction procedure of the solid material to yield antibiotic is much simpler in solid-state cultivation than in liquid cultivation, and smaller amounts of solvent are needed in solid-state cultivation. This is mainly because the water content of solid material is much lower than that of the liquid medium, and because the accumulated antibiotic concentration in solid state cultivation is higher than that in liquid cultivation. In these respects, solid-state cultivation is more promising for the mass production of *B. subtilis* as a biocontrol agent.

Plant test As a means of application of *B. subtilis* as an antagonist to plant pathogens in soil, the methods of dipping plant roots in bacterial solution (34), seed coating (70-76) and direct mixing with soil (32, 57, 77-82) have been attempted. However, these reports were rather phenomenological and there is a lack of scientific analysis for the mechanisms which are involved in the interaction with soil.

When the culture samples obtained from liquid cultiva-

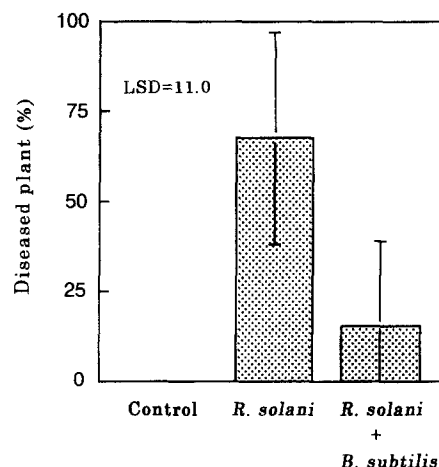


FIG. 2. Effect of product from solid-state fermentation of *B. subtilis* on the suppression of tomato disease caused by a fungal pathogen, *R. solani* in pot test. Control is a pot containing no plant pathogen. *R. solani* is a pot containing only a plant pathogen. *R. solani*+*B. subtilis* is a pot containing both a plant pathogen and the solid-state fermentation product with *B. subtilis*. LSD is a least significant difference analyzed by Fisher's statistical analysis.

tion of *B. subtilis* were applied to the biological control of crown and root rot of tomato caused by *Fusarium oxysporum*, the tomato yield was three to four times greater as compared with the control in a greenhouse test (82). The sample was also effective to reduce occurrence of bacterial wilt of tomato caused by the bacterial pathogen *Pseudomonas solanacearum*. So far, *B. subtilis* was considered to be not effective against *P. solanacearum* (83). In the soil, where occurrence of damping-off of tomato caused by *R. solani* was suppressed by the *B. subtilis* culture, cells of *B. subtilis* were recovered and antibiotics were detected (54). The product from solid-state cultivation was also applied to a plant test of tomato for the damping-off and the result is shown in Fig. 2. The occurrence of the disease was reduced to one-third of that of only the control plant pathogen. Cells of the bacterium, and the antibiotics were also recovered.

As considerable difficulty was envisaged in large-scale inoculation of the soil of a field with bacterium, the possibility of seed bacterization was explored to enhance the effectiveness of bacterial control agents (70-76).

Liberation of organic nutrients by the germinating seeds and the roots has a profound effect on the development of specific microbial populations on seed and root surfaces as in the soil around them. The soil-borne pathogens have to compete with the antagonistic microbial population of the rhizosphere in order to invade the host. The rhizosphere soils are normally known to contain higher concentrations of sugars and amino acids and thus would help in the increased production of the antibiotic, once the bacterium introduced in this region is established. The bacteria on seed would create an inhibitory zone at these loci by production of antibiotics against invasion of roots of the seedlings by the pathogens. Moreover, additional nutrients will be effective if seed bacterization is to be attempted. Seed inoculation with *B. subtilis* has decreased wilt of pigeon peas grown in wilt-sick soil, especially when the bacterial suspension was prepared with molasses and groundnut cake materials (73). The production of the antibiotic lasted for more than 10 d in soil and the amount of the pro-

duced antibiotic was higher when the seeds were bacterized in molasses and groundnut cake.

B. subtilis was used successfully to control *Pythium* and *Rhizoctonia* diseases of ornamental plants grown in soil treated by aerated steams (74). However, tests in field soils by inoculating several different strains of *B. subtilis* did not always increase the yield. The fact that the antibiotic is produced in unsterilized soil in appreciable quantities is important from the practical point of view, because most microorganisms which produce antibiotics in sterilized soil fail to do so under unsterilized conditions.

The use of biocontrol on aerial plant surfaces is much less developed than that in the soil environment. One isolate of *B. subtilis* provided control of bean rust at a level equivalent to that of following fungicide application, when *B. subtilis* and a fungicide were sprayed on plants in the greenhouse prior to inoculation with the spores of the plant pathogen (84–86).

When inconsistency and low efficiency of biocontrol occur in the field, one possible solution is the use of an integrated biocontrol system, using a mixture of several bacteria, or a mixture of chemical and biological agents. *B. subtilis* with added chemicals yielded more effective control of onion white rot caused by *Sclerotium cepivorum* (87) or root rot of field pea caused by *R. solani* (88) than chemicals alone, and the bacteria yielded season-long protection. Integrated biological control of stem canker and black scurf of potatoes caused by *R. solani* and potato scab caused by *Streptomyces scabies* by *B. subtilis* was demonstrated in both greenhouse and field experiments. In the greenhouse, the occurrence of these diseases was reduced by 60–70%, and in the field a similar effect was observed (89). The synergistic phenomenon involved in integrated control using fungicides and biocontrol agents may be more efficient and longer-lasting than the control achieved through biocontrol agents or fungicides alone.

USE OF OTHER BACTERIA

The production of organic substances such as antibiotics or siderophores plays an important role in the biological control of plant diseases as described above. However, there are relatively few reports on the production of inorganic substances that are involved in biological control. Only hydrogen cyanide, which is a secondary metabolite produced by fluorescent pseudomonads, has been extensively analyzed (90). *Alcaligenes faecalis* demonstrated the ability to suppress the growth of *Fusarium oxysporum*, via the production of a siderophore (91). However, *A. faecalis* no. 4 (no. 4), which showed a significant suppressive effect on the growth of 13 kinds of plant pathogens *in vitro*, produced hydroxylamine during heterotrophic nitrification (92). When culture broth, cell suspension and supernatant of no. 4 were applied to sterilized soil containing *R. solani*, the damping-off of tomato seedlings caused by *R. solani* was suppressed significantly. However, in unsterilized soil, damping-off was suppressed only by the culture broth and the cell suspension treatments, presumably because hydroxylamine in the supernatant was decomposed in soil. When transposon mutagenesis was applied to no. 4 to produce a non-hydroxylamine-producing mutant, this mutant showed no suppressive effect on the growth of *R. solani* either *in vitro* or *in vivo*. These data indicate

that the hydroxylamine produced by no. 4 is involved in the suppression of the plant disease.

In a natural soil system, the control of plant disease will be materialized not only by antibiosis but also in concert with other mechanisms. We expect biocontrol to contribute to the reduction of the excessive use of chemicals. Recent advances in molecular biology will provide us with a powerful tool for the application of biocontrol.

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